

Masks that calibrate themselves

Can the pixel masks of Skipper-CCD analyses do their job without being tuned to the particular sensor they run on?

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Short answer, from simulations of three different sensors (5 independent seeds). Of the 28 cases in which a mask tuned by hand on one sensor was moved unchanged to another, 6 ended up *worse than applying no mask at all* and 4 were no better than no mask. The worst was low-energy clusters on the surface laboratory sensor (tuned on deep underground), at 0.12 of the oracle where doing nothing scores 1.00.

The same masks written as procedures that calibrate themselves were worse than no mask in 1 of 14 cases (low-energy clusters on the surface laboratory sensor (adaptive), 0.96 against 1.00 for no mask), with a median 0.98 of the oracle tuned with truth on that same sensor and a worst case of 0.45 (charge-transfer trails on the surface laboratory sensor (adaptive)). Self-calibration does cost something against a mask tuned for the sensor at hand: the best transplanted mask beat it in 8 cases.

On sensors without the defect they target, the adaptive masks fired at the rate they were built for.

On the 2 seeds run once after the code was frozen and never used to change anything: 6 of 26 transplanted cases worse than no mask, 0 of 13 adaptive ones, and a median adaptive 0.98 of the oracle.

The question

A Skipper-CCD counts single electrons in millions of pixels. Before any physics is extracted, analyses discard pixels with masks, each aimed at one problem: charge left behind during transfer, hits in the serial register, hot columns and pixels, photons emitted around high-energy tracks (the halo), clusters of low-energy events, and muons. Every mask has sizes and thresholds — a radius, a trail length, a rate cut — chosen by hand for one sensor, readout and site.

The working hypothesis: what transfers between sensors is not the numbers but the *procedure* used to choose them. So each mask was written twice: a **fixed** form with hand-set constants, and an **adaptive** form that measures its own sizes on the images it is applied to and sets its threshold by a false-positive rate ($\alpha = 0.01$, corrected for the number of tests) or by measurable physical properties of the sensor.

Mask	Fixed form	Adaptive form
Hot columns & pixels	column rate above $k \times$ median	Poisson tail of each column's count against its own exposure, Bonferroni over columns, single columns before groups
Charge-transfer trails	L pixels after every bright pixel	downstream vs. upstream excess of single electrons around the same bright pixel (binomial); symmetric sources cancel
Halo	disc of radius R around bright pixels	each annulus against everything outside it (conditional binomial); no far field needed
Serial-register hits	rows with n charged pixels in a window of w	densest window of each row against the image's own occupancy, at several widths
Low-energy clusters	disc around events with $\geq m$ neighbours	neighbours within r against a Poisson count for the local valid area, at several radii
Muon tracks	charge and length cuts	charge of a minimum-ionising track for the measured length and this sensor's thickness; straight, piled-up or cut by the border

How it was tested

A simulator that remembers where every electron came from

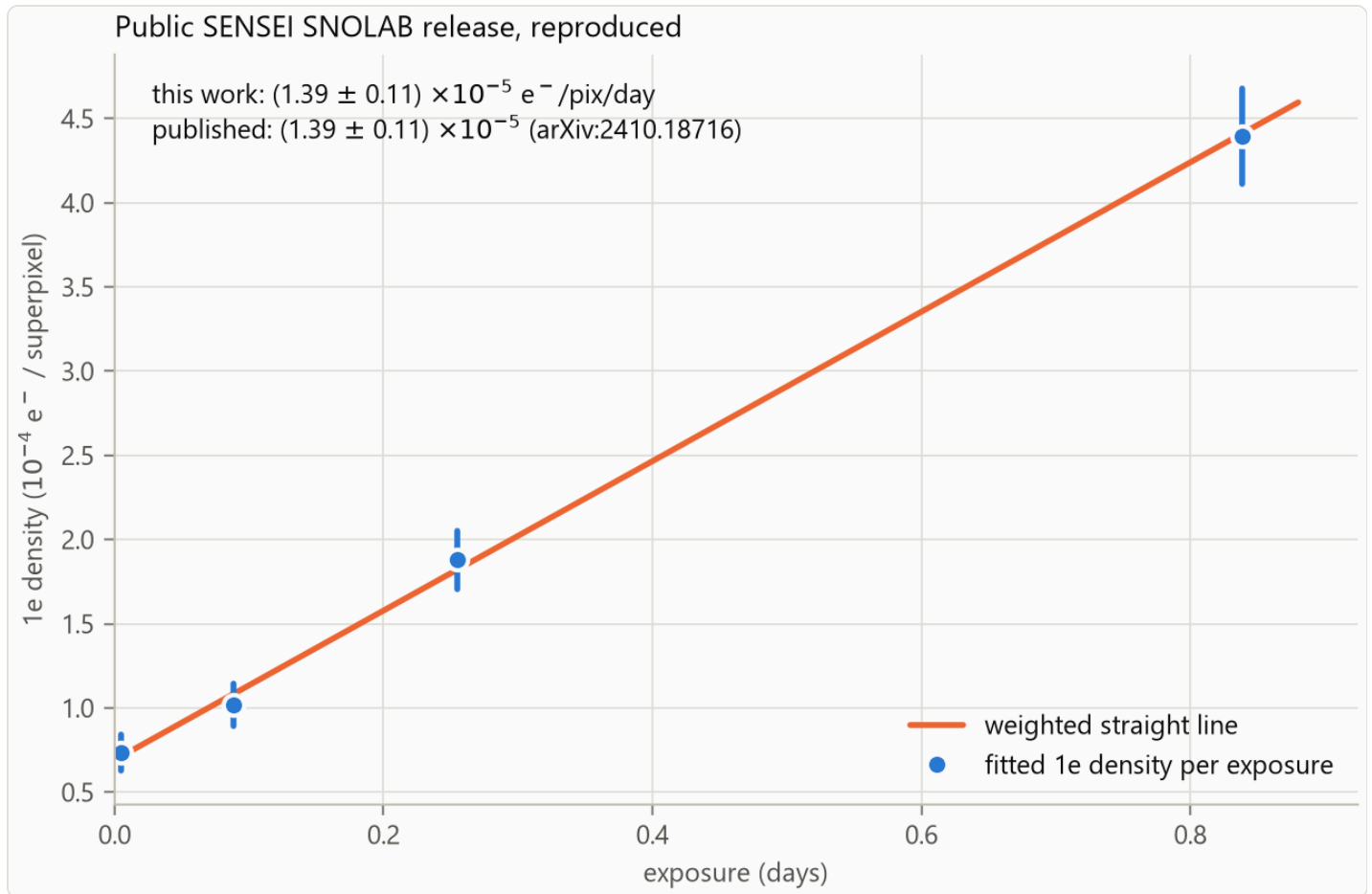
Each source of charge fills its own map, so a mask can be scored against exactly the events it should remove. Three sensors differ the way real deployments do. Their geometry, noise, dark rates, background rates and muon fluxes come from public measurements (SENSEI at SNOLAB and MINOS, Oscura sensors at the surface, SNO, PDG); defect populations are not published in a transferable form and are scenario values chosen to differ between sensors.

Sensor	Pixels	Thickness (μm)	Noise (e $^-$)	Exposure	Dark rate (e $^-$ /pix/day)	Muons (cm $^{-2}$ day $^{-1}$)	High-E (dru)	Halo μm / trail px / hot cols
Deep underground	3072 × 512	665	0.14	20 h	1.4×10^{-5}	2.9e-05	50	100 / 30 / 4
Shallow underground	3072 × 443	675	0.14	20 h	1.6×10^{-4}	6.9	3.37e+03	150 / 60 / 10
Surface laboratory	1278 × 1058	725	0.17	1 h	3.0×10^{-2}	1.4e+03	3e+04	250 / 100 / 25

Three versions of every mask were compared on an independent test stack of each sensor: the **oracle** (fixed form tuned with the simulator's truth on that sensor), the **transplant** (the oracle constants of a different sensor) and the **adaptive** form, calibrated without truth. The figure of merit is $S/\sqrt{(S+B)}$: surviving injected signal against surviving target background.

A check against real images

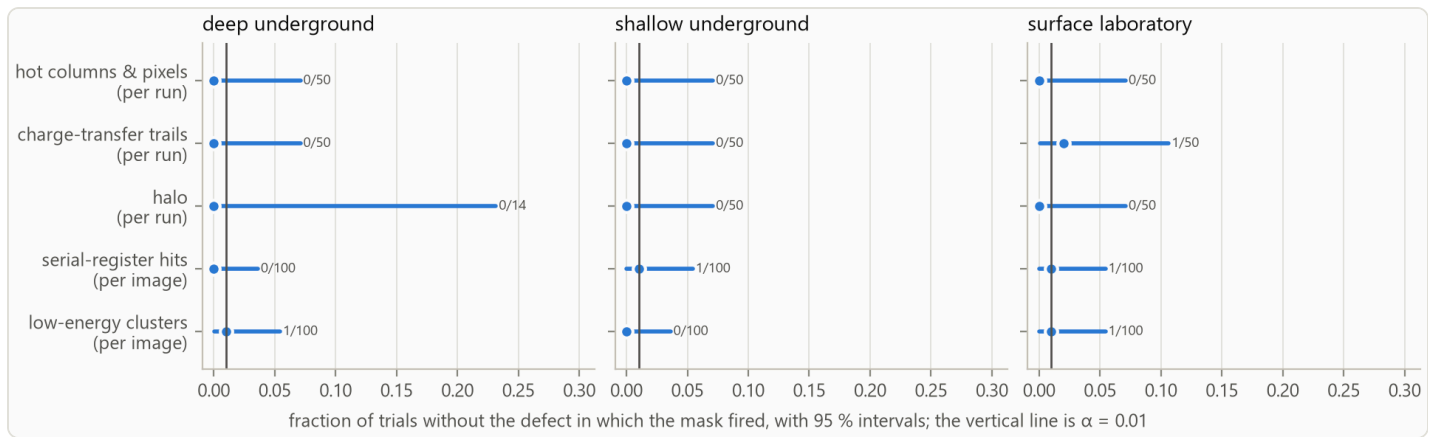
Before relying on anything, the public SENSEI SNOLAB data release was read and its single-electron rate reproduced: $1.39 \times 10^{-5} \pm 1.1 \times 10^{-6}$ e $^-$ /pix/day against the published $1.39 \times 10^{-5} \pm 1.1 \times 10^{-6}$ (pull -0.02).



Single-electron density per exposure in the public SENSEI SNOLAB release, with the weighted straight line. The slope is the rate.

Result 1 · How often the adaptive masks fire when there is nothing to find

50 independent runs of each of the three sensors with every defect the masks look for switched off, keeping what a real sensor cannot switch off: dark current, spurious charge, the injected signal, muon tracks and high-energy deposits. The stack-calibrated masks decide once per run of 2 images; the others decide per image. The test is two-sided, so a mask that is far *more* conservative than α is visible as well.



Clopper–Pearson 95 % intervals; the vertical line is $\alpha = 0.01$. Intervals are wide because 50 runs cannot resolve 0.01 from 0.03.

Sensor	Adaptive mask	Unit	Fired / trials	Rate	95 % interval	Against α
Deep underground	Hot columns	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Deep underground	Charge-transfer trails	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Deep underground	Halo	per run	0 / 14	0.000	[0.000, 0.232]	contains α
Deep underground	Serial-register hits	per image	0 / 100	0.000	[0.000, 0.036]	contains α
Deep underground	Low-energy clusters	per image	1 / 100	0.010	[0.000, 0.054]	contains α
Shallow underground	Hot columns	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Shallow underground	Charge-transfer trails	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Shallow underground	Halo	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Shallow underground	Serial-register hits	per image	1 / 100	0.010	[0.000, 0.054]	contains α
Shallow underground	Low-energy clusters	per image	0 / 100	0.000	[0.000, 0.036]	contains α
Surface laboratory	Hot columns	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Surface laboratory	Charge-transfer trails	per run	1 / 50	0.020	[0.001, 0.106]	contains α
Surface laboratory	Halo	per run	0 / 50	0.000	[0.000, 0.071]	contains α
Surface laboratory	Serial-register hits	per image	1 / 100	0.010	[0.000, 0.054]	contains α

Sensor	Adaptive mask	Unit	Fired / trials	Rate	95 % interval	Against α
Surface laboratory	Low-energy clusters	per image	1 / 100	0.010	[0.000, 0.054]	contains α

Result 2 · Transplanted constants can fail badly; self-calibration does not

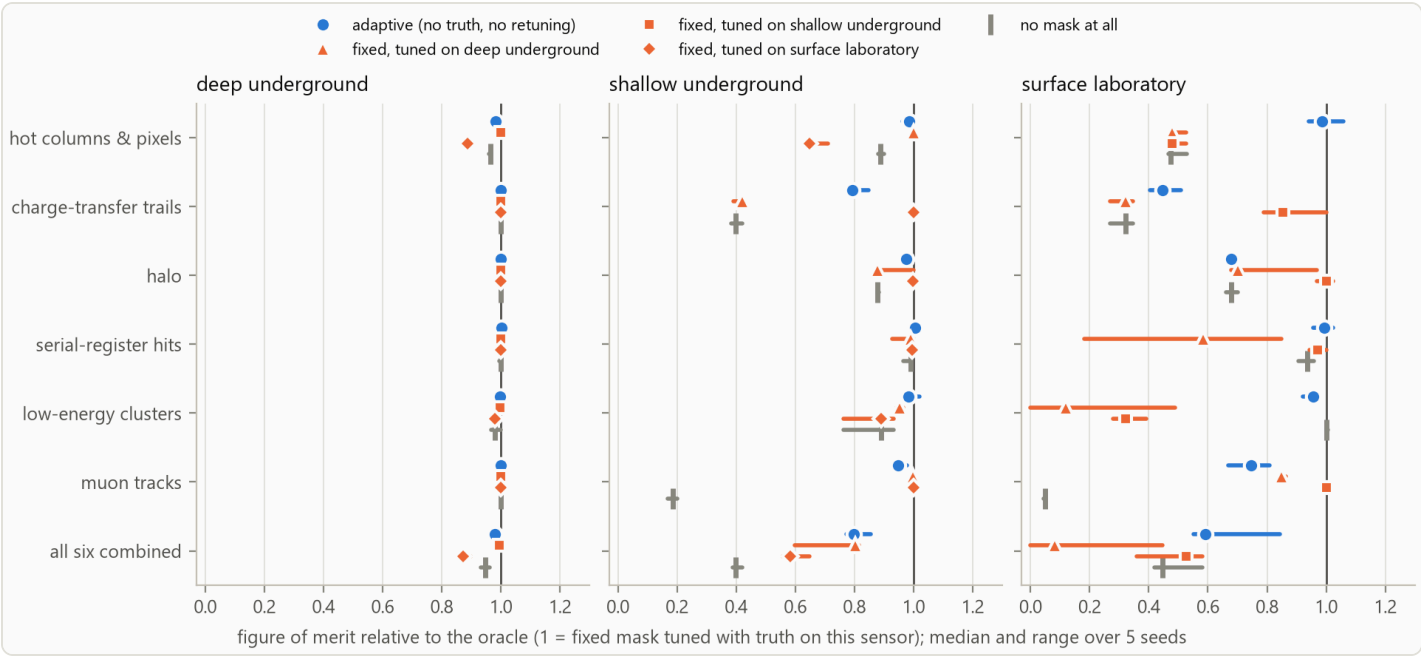


Figure of merit relative to the oracle tuned with truth on the same sensor (vertical line). Markers are medians over 5 seeds, whiskers the range. The deep-underground sensor has few target events, so every version sits near 1 there.

Evaluated on	Mask	Target events	No mask	Adaptive	Fixed, tuned on deep	Fixed, tuned on shallow	Fixed, tuned on surface
Deep underground	Hot columns & pixels	59	0.97 (0.96–0.97)	0.98 (0.98–0.99)	oracle = 1	1.00 (1.00–1.00)	0.89 (0.88–0.90)
Deep underground	Charge-transfer trails	0 too few to compare	1.00 (1.00–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	Halo	0 too few to compare	1.00 (1.00–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	Serial-register hits	3 too few to compare	1.00 (0.99–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	Low-energy clusters	25	0.98 (0.97–1.00)	1.00 (0.99–1.00)	oracle = 1	1.00 (0.99–1.00)	0.98 (0.97–1.00)
Deep underground	Muon tracks	0 too few to compare	1.00 (1.00–1.00)	1.00 (1.00–1.00)	oracle = 1	1.00 (1.00–1.00)	1.00 (1.00–1.00)
Deep underground	All six combined	81	0.95 (0.93–0.96)	0.98 (0.98–0.99)	oracle = 1	1.00 (0.99–1.00)	0.87 (0.86–0.90)
Shallow underground	Hot columns & pixels	176	0.89 (0.88–0.90)	0.98 (0.96–1.00)	1.00 (1.00–1.00)	oracle = 1	0.65 (0.63–0.71)
Shallow underground	Charge-transfer trails	3237	0.40 (0.38–0.42)	0.79 (0.78–0.85)	0.42 (0.39–0.42)	oracle = 1	1.00 (0.99–1.00)
Shallow underground	Halo	211	0.88 (0.88–0.88)	0.97 (0.97–0.99)	0.88 (0.88–1.00)	oracle = 1	1.00 (0.99–1.00)

Evaluated on	Mask	Target events	No mask	Adaptive	Fixed, tuned on deep	Fixed, tuned on shallow	Fixed, tuned on surface
Shallow underground	Serial-register hits	24	0.99 (0.96–0.99)	1.01 (1.00–1.01)	0.99 (0.93–1.00)	oracle = 1	1.00 (0.99–1.01)
Shallow underground	Low-energy clusters	176	0.89 (0.76–0.93)	0.98 (0.96–1.02)	0.95 (0.94–0.97)	oracle = 1	0.89 (0.76–0.93)
Shallow underground	Muon tracks	14116	0.19 (0.17–0.20)	0.95 (0.94–0.98)	1.00 (0.99–1.00)	oracle = 1	1.00 (1.00–1.00)
Shallow underground	All six combined	4093	0.40 (0.39–0.42)	0.80 (0.77–0.86)	0.80 (0.60–0.81)	oracle = 1	0.58 (0.56–0.65)
Surface laboratory	Hot columns & pixels	1657	0.47 (0.47–0.53)	0.99 (0.94–1.06)	0.48 (0.47–0.53)	0.48 (0.47–0.53)	oracle = 1
Surface laboratory	Charge-transfer trails	66249	0.32 (0.27–0.35)	0.45 (0.41–0.51)	0.32 (0.27–0.35)	0.85 (0.79–1.00)	oracle = 1
Surface laboratory	Halo	2049	0.68 (0.66–0.70)	0.68 (0.66–0.70)	0.70 (0.68–0.97)	1.00 (0.97–1.02)	oracle = 1
Surface laboratory	Serial-register hits	76	0.94 (0.91–0.96)	0.99 (0.96–1.02)	0.59 (0.18–0.85)	0.97 (0.94–1.00)	oracle = 1
Surface laboratory	Low-energy clusters	41	1.00 (1.00–1.01)	0.96 (0.92–0.97)	0.12 (0.00–0.49)	0.32 (0.28–0.39)	oracle = 1
Surface laboratory	Muon tracks	175494	0.05 (0.04–0.05)	0.75 (0.67–0.81)	0.85 (0.84–0.86)	1.00 (1.00–1.00)	oracle = 1

Evaluated on	Mask	Target events	No mask	Adaptive	Fixed, tuned on deep	Fixed, tuned on shallow	Fixed, tuned on surface
Surface laboratory	All six combined	70035	0.45 (0.42–0.58)	0.59 (0.55–0.84)	0.08 (0.00–0.45)	0.53 (0.36–0.58)	oracle = 1

Figure of merit relative to the oracle of the sensor in the first column; median, with the range over seeds in brackets.

What the checks caught

Every error below was found by a test or a diagnostic against the simulator's truth, fixed, and recorded.

- The simulator redrew hot columns in every image, which would have invalidated every calibration on a stack.
- The hot-column mask flagged the innocent neighbours of each hot column (17 of them in a test).
- The halo test treated its reference rate as exact; a 4σ fluctuation gave a halo where there was none.
- At the surface no pixel is far from every muon track: the first halo calibration fell back to its largest radius and masked the whole image. It now compares each annulus with everything outside it.
- Crossing muons merged into wide clusters that the track criterion rejected (24 % of muon pixels missed); tracks cut by the image border were missed too.
- Hot columns were calibrated before charge-transfer trails, whose vertical trails from muons made 69 of 81 flagged columns false.
- In one seed the halo radius, chosen by significance alone, reached 125 px on the surface sensor and masked 99.9 % of the image. The radius now maximises a figure of merit estimated from the data within the significant range; all seeds were re-run, plus two seeds never used during development.

Limitations

- Defect populations (hot columns, trail lengths, serial hits, clusters) are scenario values, not measurements.
- One figure of merit; the oracle is tuned mask by mask, not jointly, so a combined adaptive mask can exceed it.

- Adaptive trail lengths are limited by the number of bright pixels in the calibration stack: they are short when data are few.
- The muon figure of merit counts pixels, which weighs every missed track pixel heavily.
- An adaptive mask cannot tell injected signal from dark current, so it estimates signal as every uniform single electron; the evaluation counts only the injected signal. At the surface, where dark current is ~ 20 times the signal, the adaptive halo therefore chooses not to mask and reaches ~ 0.7 of the oracle. The evaluation was fixed before this was seen and was not changed.
- The adaptive masks were not yet run on the real public images beyond the rate reproduction.

Reproduce

```
conda env create -f environment.yml
conda activate skmask
python scripts/fetch_public_data.py
pytest
python analysis/reproduce_release_rate.py
python analysis/check_release_mask_bits.py
python analysis/null_false_positive_rates.py
python analysis/compare_masks_across_sensors.py 4 --seed 20260915
python analysis/figure_compare_masks.py
python analysis/build_report.py
python scripts/make_pdf.py
```

Every output has a `.provenance.json` sidecar with the script, git commit, input hashes, parameters and seed; [PROVENANCE.md](#) says how each result was checked.